

EXERCISE 2 SQL Aggregate Functions & Operators

01 The students Table

QUESTION 1 List all distinct departments in the students table.

```
--Students table questions
--QUESTION 1 01 List all distinct departments in the students table
SELECT DISTINCT department
FROM excercise2.learnbright.students;
```

Add parameter

Table ▾ +		🔍 ⚙️ 📄 🗑️ ^ ✕					
	department						
1	IT						
2	HR						
3	Finance						

QUESTION 2 Get the average age of students per department.

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```
--QUESTION 2 Get the average age of students per department.
```

```
SELECT department, AVG(age) AS avg_age
```

```
FROM excercise2.learnbright.students
```

```
GROUP BY department;
```

Add parameter

	A _c department	1.2 avg_age
1	IT	20.5
2	HR	22
3	Finance	23

Question 3 Show departments with more than 1 student.

```
--Question 3 Show departments with more than 1 student
SELECT department, COUNT(*) AS student_count
FROM excercise2.learnbright.students
GROUP BY department
HAVING COUNT(*) > 1;
```

Add parameter

Table  

  

	 department	 student_count
1	IT	2
2	HR	2

Question 4 Get all students whose age is between 21 and 23

```
--Question 4 Get all students whose age is between 21 and 23.  
SELECT *  
FROM excercise2.learnbright.students  
WHERE age BETWEEN 21 AND 23;
```

Add parameter

Table

	student_id	name	age	department
1	2	Bob	22	HR
2	3	Charlie	21	IT
3	4	Diana	23	Finance
4	5	Eve	22	HR

4 rows

0.627s runtime

[See performance \(1\)](#)

✓ Checked for improvements

Refreshed Now

Question 5 List all students in the IT or HR department who are older than 21.

```
--Question 5 List all students in the IT or HR department who are older than 21.
SELECT *
FROM excercise2.learnbright.students
WHERE department IN ('IT', 'HR')
AND age > 21;
```

Add parameter

Table ⌵ + 🔍 🏠 📄 📊 ⌵ ✕

	¹ ₃ student_id	^A _C name	¹ ₃ age	^A _C department
1	2	Bob	22	HR
2	5	Eve	22	HR

Question 6 Show total credits per department, only for departments with more than 5 total credits.

▶ Run selected (1000) ✓ Just now (<1s) workspace: default

Edit

```
--QUESTION 6 Show total credits per department, only for departments with more than 5 total credits
SELECT department, SUM(credits) AS total_credits
FROM excercise2.learnbright.courses
GROUP BY department
HAVING total_credits > 5;
```

Add parameter

Table +

🔍 🏠 📄 📊 ⌵ ✕

	^A _C department	¹ ₃ total_credits
1	IT	11

⬇ ⌵

1 row | 0.429s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed Now

Question 7 List all courses that do not have 4 credits.

```
--QUESTION 7 List all courses that do not have 4 credits
SELECT DISTINCT course_id,
       .....course_name,
       .....department,
       .....credits
FROM   excercise2.learnbright.courses
WHERE  credits <>4;
```

Add parameter

Table  

      

	¹ ₃ course_id	^A _C course_name	^A _C department	¹ ₃ credits
1	105	Statistics	HR	3
2	101	SQL Basics	IT	3
3	104	Exel	Finance	2

Question 8 Show the top 3 courses by credits in descending order.

--QUESTION 8 Show the top 3 courses by credits in descending order

SELECT DISTINCT course_id,
course_name,
credits
FROM excercise2.learnbright.courses
ORDER BY credits DESC
LIMIT 3;

Add parameter

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Q

Y

^

X

	123 course_id	A0 course_name	123 credits
1	103	Data Science	4
2	102	Python	4
3	101	SQL Basics	3

Question 9 Get the maximum, minimum, and average grade across all enrollments.

--QUESTION 9 Get the maximum, minimum, and average grade across all enrollments

SELECT MAX(grade) AS max_grade,
MIN(grade) AS min_grade,
AVG(grade) AS avg_grade
FROM excercise2.learnbright.enrollments;

Add parameter

Table +

1 2 3 max_grade1 2 3 min_grade1 2 avg_grade

1907884.6

↓

▼

1 row

0.325s runtime

[See performance \(1\)](#)

✓ Checked for improvements

Refreshed Now

Question 10 Count how many enrollments exist per course.

--QUESTION 10 Count how many enrollments exist per course

SELECT course_id,
*****COUNT(*) AS enrollment_count
FROM excercise2.learnbright.enrollments
GROUP BY course_id;

Add parameter

Table +

Q Y I □ ⌂ ^ X

	^{1 2} 3 course_id	^{1 2} 3 enrollment_count
1	101	1
2	102	1
3	103	1
4	104	1
5	105	1

↓

▼

5 rows | 0.397s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed Now

Question 11 Find total salary and total bonus per department

```
-- QUESTION 11 Find total salary and total bonus per department
SELECT department,
       SUM(salary) AS total_salary,
       SUM(bonus) AS total_bonus
FROM excercise2.learnbright.salaries
GROUP BY department;
```

Add parameter

Table  

     

	department	total_salary	total_bonus
1	IT	122000	10500
2	HR	109000	7500
3	Finance	70000	6000

 3 rows | 0.417s runtime [See performance \(1\)](#)  Checked for improvements

Refreshed Now

Question 12 Show departments where average salary is above 55,000.

--QUESTION 12 Show departments where average salary is above 55,000

SELECT department,

.....AVG(salary) AS avg_salary

FROM excercise2.learnbright.salaries

GROUP BY department

HAVING avg(salary)>55000;

Add parameter

Table

+

Q F L □ ⌂ ^ X

	A ^B _C department	1.2 avg_salary
1	IT	61000
2	Finance	70000

↓

▼

2 rows

0.335s runtime

[See performance \(1\)](#)

✓ Checked for improvements

Refreshed Now

Question 13 List employees whose salary plus bonus is greater than 60,000

```
--QUESTION 13 List employees whose salary plus bonus is greater than 60,000
SELECT
  employee_id,
  name,
  salary,
  bonus,
  (salary+bonus) AS total_compensation
FROM excercise2.learnbright.salaries
WHERE (salary+bonus)>60000;
```

Add parameter

Table +

Q F L W R ^ X

	employee_id	name	salary	bonus	total_compensation
1	1	TOM	60000	5000	65000
2	3	Spike	70000	6000	76000
3	4	Tyke	62000	5500	67500

↓ ▾

3 rows | 0.826s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed Now

Question 14 Show total and average budget per department. Only include departments with average budget above 70,000.

```
--QUESTION 14 Show total and average budget per department. Only include departments with average budget above 70,000
SELECT
  department,
  SUM(budget) AS total_budget,
  AVG(budget) AS avg_budget
FROM excercise2.learnbright.projects
GROUP BY department
HAVING AVG(budget)>70000;
```

Add parameter

Table +

1.2 avg_budget

1.2 avg_budget

	A department	1.2 total_budget	1.2 avg_budget
1	IT	270000	135000
2	Finance	80000	80000

↓

2 rows

1.241s runtime

[See performance \(1\)](#)

✓ Checked for improvements

Refreshed Now

Question 15 List all projects with budgets between 50,000 and 120,000, excluding the Marketing department.

```
--QUESTION 15 List all projects with budgets between 50,000 and 120,000, excluding the Marketing department
SELECT
  project_id,
  project_name,
  department,
  budget
FROM excercise2.learnbright.projects
WHERE budget BETWEEN 50000 AND 120000
AND department <> 'Marketing';
```

Add parameter

Table +

Q F I □ ⌂ ^ X

	project_id	project_name	department	budget
1	1	AI App	IT	120000
2	2	Payroll System	Finance	80000
3	5	HR Portal	HR	50000

↓ ▾

3 rows | 0.420s runtime [See performance \(1\)](#) ✓ Checked for improvements

Refreshed Now

